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Evolution of microstructure, hardness, and corrosion properties of high-entropy $Al_{0.5}CoCrFeNi$ alloy

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In this study, we investigate the microstructure, hardness, and corrosion properties of as-cast $Al_{0.5}CoCrFeNi$ alloy as well as $Al_{0.5}CoCrFeNi$ alloys aged at temperatures of 350 °C, 500 °C, 650 °C, 800 °C and 950 °C for 24 h. The microstructures of the various specimens are investigated using X-ray diffraction (XRD), scanning electron microscopy (SEM), and electron probe X-ray microanalysis (EPMA). The results show that the microstructure of as-cast $Al_{0.5}CoCrFeNi$ comprises an FCC solid solution matrix and droplet-shaped phases (Al–Ni rich phases). At aging temperatures of between 350 and 950 °C, the alloy microstructure comprises an FCC + BCC solid solution with a matrix, droplet-shaped phases (Al–Ni rich phase), wall-shaped phases, and needle-shaped phases (Al–(Ni, Co, Cr, Fe) phase). The aging process induces a spinodal decomposition reaction which reduces the amount of the Al–Ni rich phase in the aged microstructure and increases the amount of the Al–(Ni, Co, Cr, Fe) phase. The hardness of the $Al_{0.5}CoCrFeNi$ alloy increases after aging. The optimal hardness is obtained at aging temperatures in the range 350–800 °C, and the hardening effect decreases at higher temperatures. Both the as-cast and aged specimens are considerably corroded when immersed in a 3.5% NaCl solution because of the segregation of the Al–Ni rich phase precipitate formed in the FCC matrix. Cl^- ions preferentially attack the Al–Ni rich phase, which is a sensitive zone exhibiting an appreciable potential difference, with consequent galvanic action.

1. Introduction

Many practical alloy systems have been developed over the years for a wide range of engineering and industrial applications. Most of these alloy systems are based on one or two dominant elements, e.g. ferrous alloys, intermetallic alloys [1,2], bulk metallic glass alloys [3–5], and so on. However, according to conventional metallurgical theory, the presence of multiple elements in an alloy system results in the formation of various compounds, which give rise to a complex microstructure with poor properties. Consequently, Yeh et al. [6,7] developed high-entropy alloy systems (HEASs), in which the multiple elements were added in equi-molar or near equi-molar ratios. Such systems have a far broader range of applications than conventional alloys and may contain five or even more major alloying elements at levels of 5–35 at.% [8–12]. Solid solution structures with multiple principal elements are more stable than conventional FCC, BCC, or FCC + BCC structures since they form nano-precipitates in the as-cast state on account of their large entropies of mixing [6,11]. HEASs have excellent high-temperature properties, mechanical properties, and corrosion resistance, and they are therefore ideally suited to industrial applications. In practice, the properties of a HEAS depend on the relative compositions of the various elements within the system [13–20]. The microstructure and properties of the $Al_xCoCrFeNi$ alloy, a typical HEAS, have been extensively studied and reported in the literature [14,21–23]. Moreover, the effects of the addition or substitution of various constituents of $Al_xCoCrFeNi$ alloy on the microstructure and properties of the alloy system at room temperature or elevated temperatures have been examined [13,24,25]. However, the hardness, corrosion properties, and microstructure evolution of $Al_xCoCrFeNi$ alloy over a range of aging temperatures have not been sufficiently studied. Accordingly, this study examines the microstructure, hardness, and corrosion properties of as-cast $Al_{0.5}CoCrFeNi$ alloy as well as $Al_{0.5}CoCrFeNi$ alloys aged at temperatures of 350 °C, 500 °C, 650 °C, 800 °C, and 950 °C, respectively, for 24 h.

2. Experimental procedures

The high-entropy $Al_{0.5}CoCrFeNi$ alloy system was melted under an Ar atmosphere in an arc furnace with adequate amounts of

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highly pure Al, Co, Ni, Cr, Fe, and Ni elements (99.99%). Specimens with dimensions of 10 mm × 10 mm × 7 mm were prepared. Some of the specimens were retained in an as-cast condition, whereas the remaining specimens were aged in an Ar atmosphere for 24 h at temperatures of 350 °C, 500 °C, 650 °C, 800 °C or 950 °C and then quenched in water. The crystalline phases of the various alloy specimens were identified via X-ray diffraction (XRD, Rigaku D/Max-2500 diffractometer with Cu-Kα radiation) at a scanning speed of 1°/min, voltage of 30 kV, and current of 50 mA. The microstructures and chemical compositions of the alloy were then examined using energy dispersive X-ray spectrometry (EDS) and electron probe X-ray microanalysis (EPMA, JEOL JXA-8009R) techniques. The hardness of the as-cast and aged specimens was measured using a Vickers hardness tester (Matsuzawa, Seiki MV-1) under a load of 300 kgf for 15 s. Finally, the electrochemical corrosion properties of the various specimens when immersed in a 3.5% NaCl solution at room temperature were determined from the anodic polarization curves obtained using a potentiodynamic method. Each specimen was scanned potentiodynamically at a rate of 1 mV s^{-1} from an initial potential of 0.5 V_{SCE} with reference to the open circuit potential to a final potential of −1.5 V_{SCE}. Note that calibration was performed in accordance with the ASTM G59 test standard [26].

3. Results and discussion

3.1. Hardness of $Al_{0.5}$CoCrFeNi alloy

Fig. 1 shows the hardness values of the $Al_{0.5}$CoCrFeNi HEAS specimens aged at different temperatures. The hardness increases from H_V 247 in the as-cast condition to between H_V 256 and H_V 285 following aging at temperatures in the range of 350–950 °C. In addition, maximum age hardening occurs at temperatures between 350 °C and 800 °C, i.e. the hardness reduces as the temperature is further increased to 950 °C. The age hardening phenomena are discussed later in relation to the microstructural observations.

3.2. X-ray diffraction of $Al_{0.5}$CoCrFeNi alloy

Fig. 2 shows the XRD patterns of the as-cast and aged specimens. The results show that the as-cast specimen is an FCC solid solution, whereas the aged specimens are FCC + BCC solid solutions. The lattice constants of the BCC and FCC solid solutions are calculated from the principal peaks to be 2.834 Å and 3.562 Å,

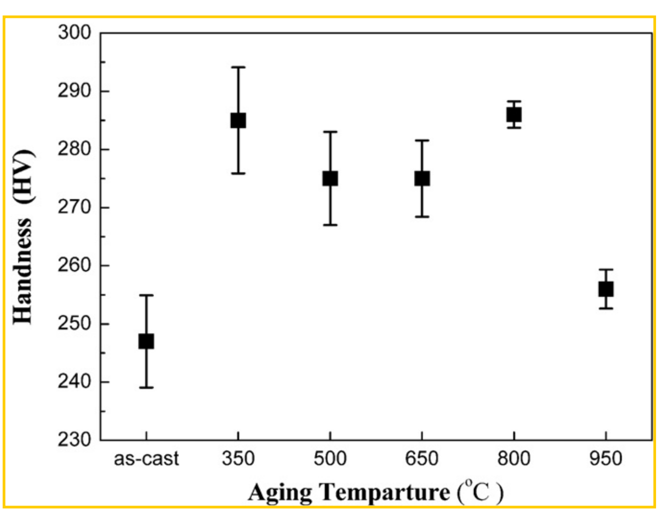

Fig. 1. Hardness of $Al_{0.5}$CoCrFeNi alloy processed at various aging temperatures.

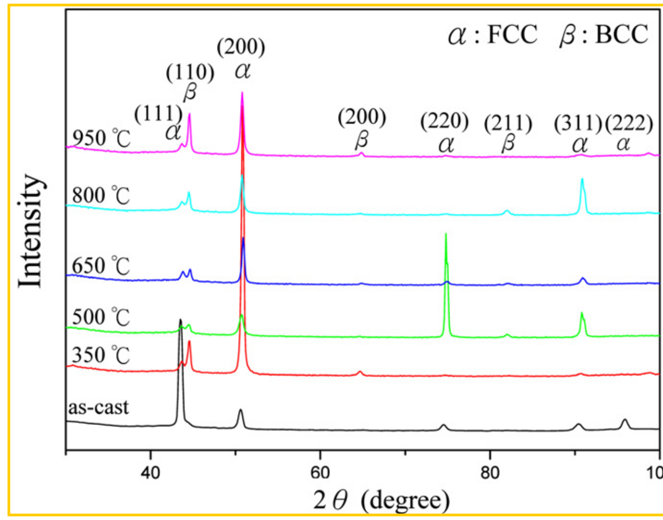

Fig. 2. XRD patterns of $Al_{0.5}$CoCrFeNi alloy processed at various aging temperatures.

respectively. As the aging temperature is increased from 350 to 950 °C, the FCC peak intensity approaches that of the BCC peak at $2\theta = 42–48°$. As the aging temperature is increased further, the FCC peak intensity decreases, whereas the BCC peak intensity increases. Thus, it is inferred that aging at high temperatures causes the $Al_{0.5}$CoCrFeNi alloy to gradually transform from a stable FCC structure to a BCC structure. From an overall perspective, Figs. 1 and 2 suggest that the transition of the $Al_{0.5}$CoCrFeNi alloy system to a BCC solid solution increases the hardness of the alloy on account of the corresponding lattice distortion.

3.3. Microstructural characteristics of $Al_{0.5}$CoCrFeNi alloy

Fig. 3 shows backscattered electron images (BEI) of the as-cast and aged specimens. In general, the BEI images show that in addition to the matrix, the solidified microstructures contain a droplet-shaped phase (DSP; in every specimen) and wall-shaped and needle-shaped phases (WSP and NSP, respectively; in the aged specimens only). As discussed above, the as-cast specimen has an FCC solid solution structure, whereas the aged specimens have an FCC + BCC solid solution structure. Table 1 lists the chemical compositions of the various as-cast and aged specimens as obtained from the EDX analysis results. The results show that the matrix and WSPs are Al-depleted phases, whereas the DSPs are Al–Ni rich phases. Furthermore, the size of the DSPs reduces as the aging temperature is increased, and NSP precipitates rich in Al as well as Ni, Co, Cr, and Fe are formed in the matrices of all the aged specimens, as shown in Fig. 3(b)–(f). In general, the BEI images suggest that the Al content of the $Al_{0.5}$CoCrFeNi alloy system prompts the formation of a BCC solid solution structure under thermal effects. Note that this inference is consistent with the findings of previous studies [6,21,27]. The formation of a considerable amount of intragranular phase segregates occurs at high aging temperatures. The segregation ratio (SR) represents the degree of element segregation and is defined as

$$SR = \frac{\text{MP in zone A}}{\text{DSP, WSP, or NSP in zone B}},$$

where zone A is the zone of the matrix phase (MP); zone B, the zone of segregation of the DSP, WSP, or NSP.

Table 3 lists the phase SRs. The matrix phases are the major phases of the as-cast specimens and of those aged at up to 950 °C,

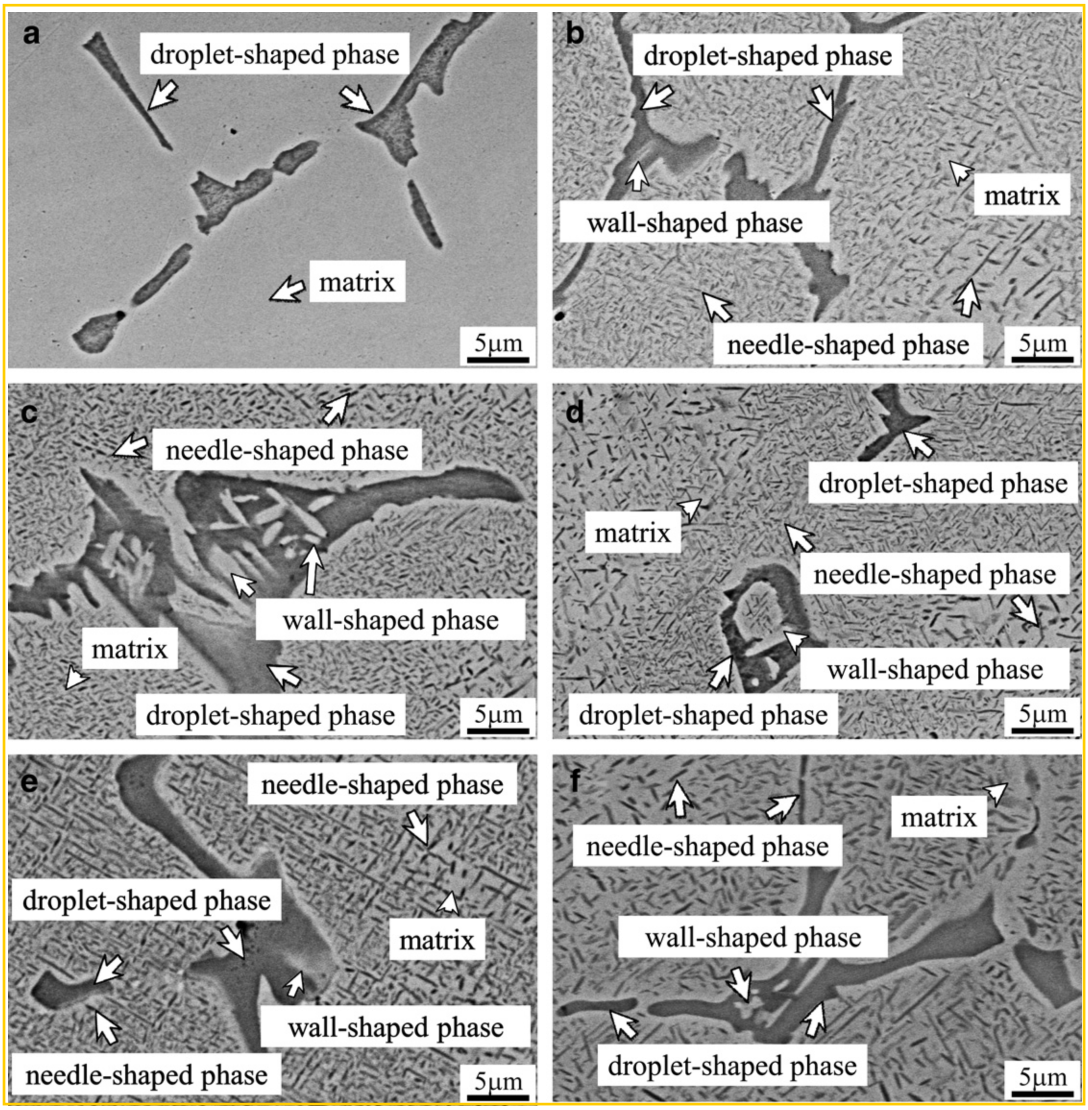

Fig. 3. BEI images of $Al_{0.5}CoCrFeNi$ alloy specimens processed at various aging temperatures: (a) as-cast, (b) 350 °C, (c) 500 °C, (d) 650 °C, (e) 800 °C, and (f) 950 °C.

The matrix phase becomes an NSP when the specimen is treated from 350 to 950 °C. The BEI images in Fig. 3 show that the FCC solid solution structure (i.e. the DSPs) and the BCC solid solution phases (i.e. the NSPs) segregate from the FCC matrix at elevated aging temperatures. This phase segregation effect occurs because the Al–Ni rich phase and the Al–(Ni, Co, Cr, Fe) phases have a higher negative mixing enthalpy than the other atom pairs of the five principal elements in the alloy system (Table 2). In practice, the negative mixing enthalpy of these particular atoms means that it is virtually impossible to prevent the segregation of the Al–Ni rich phase and the Al–(Ni, Co, Cr, Fe) phases during heat treatment [28–30]. In addition, the specimens aged at temperatures between

Table 1
EDX analysis results for chemical composition (at.%) of Al0.5CoCrFeNi alloy samples processed at different aging temperatures.

		Unit: at.%							Unit: at.%				
		Al	Co	Cr	Fe	Ni			Al	Co	Cr	Fe	Ni
As-cast	NSP	–	–	–	–	–	650 °C	NSP	15	21	21	23	22
	WSP	–	–	–	–	–		WSP	17	20	21	19	23
	DSP	25	19	12	12	32		DSP	26	18	10	12	34
	MP	10	23	22	23	22		MP	8	23	21	24	24
350 °C	NSP	15	22	19	19	25	800 °C	NSP	14	23	19	23	21
	WSP	17	21	19	20	24		WSP	16	21	18	20	25
	DSP	24	17	12	14	33		DSP	25	16	12	12	35
	MP	8	24	22	25	22		MP	10	23	21	22	24
500 °C	NSP	17	21	19	21	22	950 °C	NSP	19	20	19	21	21
	WSP	18	20	19	19	23		WSP	15	21	25	24	18
	DSP	21	20	14	16	29		DSP	26	17	7	12	38
	MP	9	24	21	23	23		MP	9	24	21	23	23

Table 2
Chemical mixing enthalpy of atom pairs in $Al_{0.5}$CoCrFeNi alloy system

Element	Unit: KJ/mol				
	Fe	Co	Ni	Cr	Al
Fe	–	−1	−2	−1	−11
Co		–	0	−4	−19
Ni			–	−7	−22
Cr				–	10
Al					–

350 and 950 °C contained WSPs within the DSPs, as shown in Fig. 3(b)–(f). The belief is that the formation of these WSPs was the result of the metastable phase. The presence of the WSPs could be ascribed to the rapid cooling of each component to room temperature, wherein the component began solidifying into its preferred crystalline structure, and to the setting of the metastable structure established by aging at temperatures between 350 and 950 °C. Accordingly, the composition of various elements and atoms differed during the diffusion process. In the DSP structure, the cooling rate of the WSPs was very high due to of the small volume (i.e. the WSPs). As a result, the diffuse field of solutes in the liquid was in a state of non-equilibrium, and the concentration and flux of solutes could not be predicted using classical local equilibrium theory. These effects caused the shortening of the mean free path of diffusing atoms, which in turn led to the formation of a metastable structure.

As discussed above, NSPs are precipitated out of the matrix during thermal aging at 350–950 °C; this is because the mixing enthalpy of the Al–Ni rich phase and the Al–(Ni, Co, Cr, Fe) phases is lower (i.e. more negative) than that of the other atom pairs of the five principal alloying elements [14]. Because the growth of the NSPs follows a parabolic kinetic equation, it can be considered that their growth is diffusion-controlled [31]. Therefore, the aging temperature has a significant effect on the growth rate of the NSPs. Thus, the size of the NSPs increases with temperature, as shown in Fig. 3(b)–(f). Further study showed that the DSP appears because of a relatively high degree of negative mixing enthalpy between Al and Ni, which tend to form atom pairs and segregate within the matrix. In other words, at first, the phase has a precipitate structure in the earliest stages of as-cast. With an increase in temperature, these DSPs grow in size and decrease in number, with a nonuniform distribution. Following thermal aging between 350 and 950 °C, the NSP could be attributed to segregation of heterogeneous atoms resulting in the microstructure of DSPs adjacent to the substrate. This could be described as large quantities of needle-like precipitate sprinkled throughout the matrix. When the temperature increased, the NSPs were distributed densely throughout the matrix and began transforming then the volume fraction of NSPs (i.e. after aging treatment) gradually increased. These NSPs are often called secondary precipitation structures. The NSPs are considered to be directly related to the age hardening effect shown in Fig. 1. At lower aging temperatures, the DSPs (Al–Ni rich phase) do not decompose, and tiny NSPs (Al–(Ni, Co, Cr, Fe) phase) are precipitated in the matrix. These highly dense precipitates strongly influence the hardening process. At temperatures greater than 950 °C, the NSPs have a larger size and are therefore more sparsely distributed in the matrix. As a result, the hardness of the aged microstructure reduces (Fig. 1).

Fig. 4 shows BEI images and the corresponding EPMA elemental mapping (without etching) results for the as-cast specimen and for two specimens aged at temperatures of 650 and 950 °C. The results show that the DSPs are composed primarily of Al and Ni, whereas the matrix is composed mainly of Co, Fe, Cr, and Ni. The difference in the composition of the two structures is the result of the non-equilibrium solidification conditions. In the DSPs of the as-cast specimen (Fig. 4(a)), the outward diffusion of the Co, Fe, and Cr elements occurs more rapidly than the inward diffusion of Al and Ni. However, as the aging temperature is increased, the rate of inward diffusion of Al and Ni increases. Thus, as shown in Fig. 4(b) and (c) for aging temperatures of 650 and 950 °C, respectively, diffusion into the DSPs is dominated by the Al and Ni elements.

These experimental results and reasonable inferences regarding the factors based on the temperature of aging and the rate of element diffusion. When the as-cast specimen is prepared, the molten components are transferred individually to the liquid alloy and are mixed thoroughly by thermal agitation. As a result, no matrix or major component can be dominant in the solidification process because the atoms of each component are solutes in the alloy system. As the temperature of the specimen reduces, each component begins to solidify into its preferred crystalline structure. Finally, as the alloy solidifies, the atoms of the various elements interfere with one other during the diffusion process [7]. In the molten state, the elements are not in equilibrium and the concentration and flux of the solutes cannot be predicted. The effects of radius, crystalline structure, and melting point shorten the mean free path of the diffusing atoms and lead to the formation of the precipitated phase. Previous studies have indicated that multicomponent alloys form complex precipitate phases through affinity reactions, whereby atoms diffuse during metal solidification [18,31]. The Al–Ni rich phase in the as-cast specimen shown in Fig. 4(a) is formed since Al has a higher affinity to Ni than to any of the other elements in the $Al_{0.5}$CoCrFeNi alloy system [32,33]. The amount of Al–Ni rich phase in the aged specimens reflects the competing effects of the affinity between the Al and Ni atoms and the diffusion force of the Ni atoms, and it depends on the aging temperature. The affinity between Al and Ni favours the formation of Al–Ni rich phases, whereas the diffusion force of the Ni atoms undoubtedly acts to weaken it. At an aging temperature of 650 °C, the formation of the Al–Ni phase is dominated by the affinity between the Al and Ni atoms since the droplet-shaped structures are thin; thus, the Ni diffusion force is relatively low. However, at a higher temperature of 950 °C, the interaction effect between Al

Table 3
Segregation ratios (SRs) for droplet-shaped phase (DSP), wall-shaped phase (WSP) and needle-shaped phase (NSP).

		Al	Co	Cr	Fe	Ni
As-cast	SR_1	0.4	1.2	1.8	1.9	0.7
	SR_2	–	–	–	–	–
	SR_3	–	–	–	–	–
350 °C	SR_1	0.5	1.1	1.2	1.3	0.9
	SR_2	0.5	1.1	1.2	1.3	0.9
	SR_3	0.3	1.4	1.8	1.8	0.7
500 °C	SR_1	0.5	1.1	1.1	1.1	1.0
	SR_2	0.5	1.2	1.1	1.2	1.0
	SR_3	0.4	1.2	1.5	1.4	0.8
650 °C	SR_1	0.5	1.1	1.0	1.0	1.1
	SR_2	0.5	1.2	1.0	1.3	1.0
	SR_3	0.3	1.3	2.1	2.0	0.7
800 °C	SR_1	0.7	1.0	1.1	1.0	1.1
	SR_2	0.6	1.1	1.2	1.1	1.0
	SR_3	0.4	1.4	1.8	1.8	0.7
950 °C	SR_1	0.5	1.2	1.1	1.1	1.1
	SR_2	0.6	1.1	0.8	1.0	1.3
	SR_3	0.3	1.4	3.0	1.9	0.6

$SR_1 = \frac{\text{Matrix phase (MP)}}{\text{Droplet} - \text{shaped phase (DSP)}}$.

$SR_2 = \frac{\text{Matrix phase (MP)}}{\text{Wall} - \text{shaped phase (WSP)}}$.

$SR_3 = \frac{\text{Matrix phase (MP)}}{\text{Needle} - \text{shaped phase (NSP)}}$.

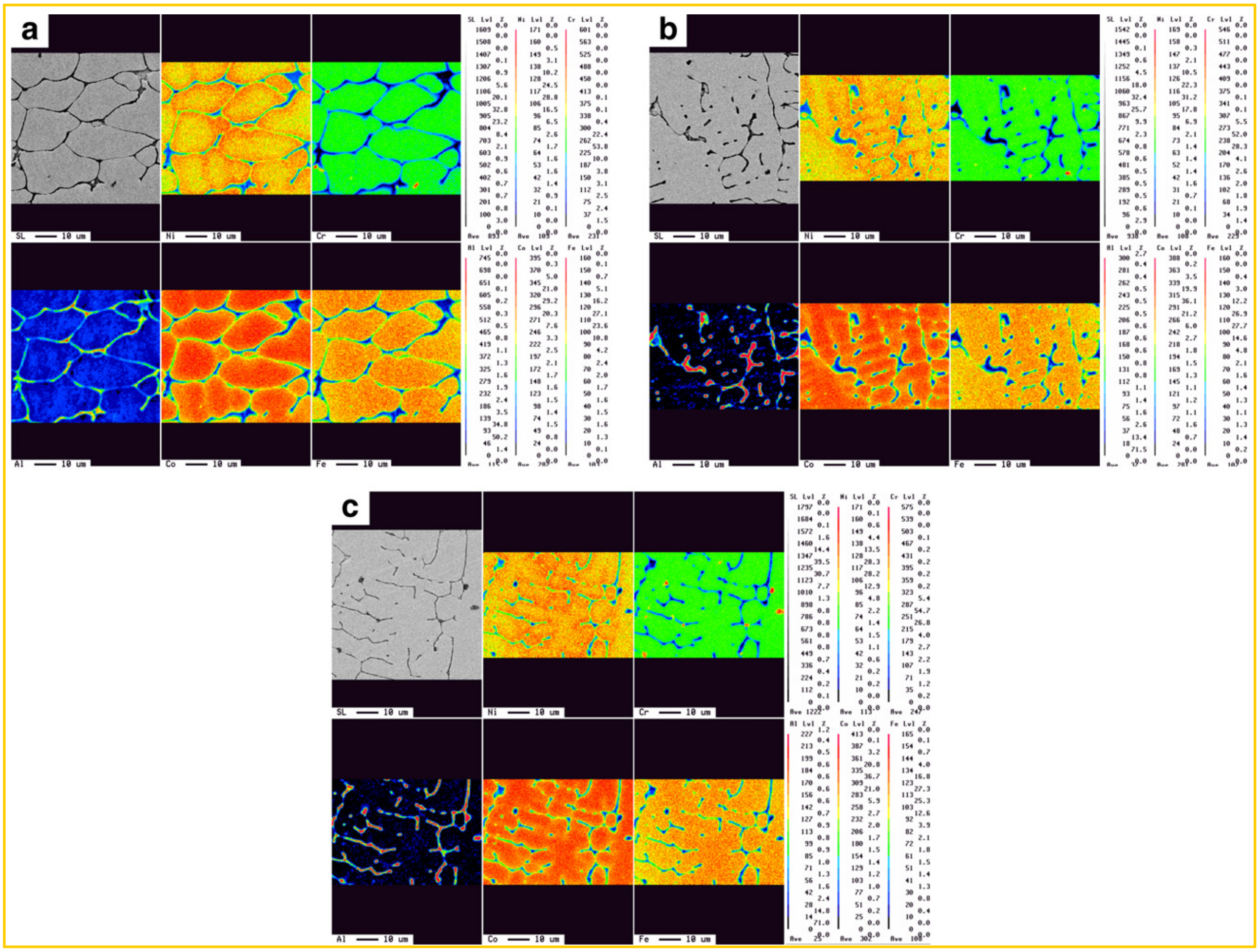

Fig. 4. BEI images and EPMA elemental mapping results for $Al_{0.5}CoCrFeNi$ alloy processed at various aging temperatures: (a) as-cast, (b) 650 °C, and (c) 950 °C.

and Ni is outweighed by the Ni diffusion force. In other words, the formation of the Al–Ni phase is dominated by the effects of Ni diffusion. In general, the Al and Ni atoms diffuse more rapidly at a higher aging temperature. As a result, the size of the Al–Ni rich droplet structures decreases with an increase in the aging temperature. The images shown in Fig. 4(b) and (c) suggest that the droplet structures attain their maximum size and the Al and Ni contents reach their respective saturation levels at aging temperatures higher than 650 °C.

3.4. Corrosion properties of $Al_{0.5}CoCrFeNi$ alloy

The electrochemical corrosion properties of the as-cast and aged specimens when immersed in a 3.5% NaCl solution were evaluated using a potentiodynamic polarization method. The polarization curves for the various alloy specimens are shown in Fig. 5, together with that of AISI 304L stainless steel for comparison purposes. The measured values of the electrochemical parameters of the AISI 304L stainless steel and alloy specimens are listed in Table 4. The AISI 304L stainless steel specimen exhibits a corrosion potential of -0.22 V_{SCE}. All of the as-cast and aged specimens have corrosion potentials in the range from 0.04 to -0.64 V_{SCE} and pitting potentials in the range from 0.13 to 0.71 V_{SCE}. Meanwhile, the as-cast and aged specimens have corrosion current densities in the range of 1.58×10^{-6}–5.29×10^{-7} A/cm^2, whereas the AISL 304L stainless steel specimen has a corrosion current density of 6.00×10^{-8} A/cm^2. The corrosion rates of the alloy specimens aged at higher temperatures are significantly greater than those of the specimens aged at lower temperatures. However, the corrosion properties of the alloy aged at 650 °C are notably different from those of the other aged specimens. This difference is most probably the result of the microstructural transformation induced by the aging

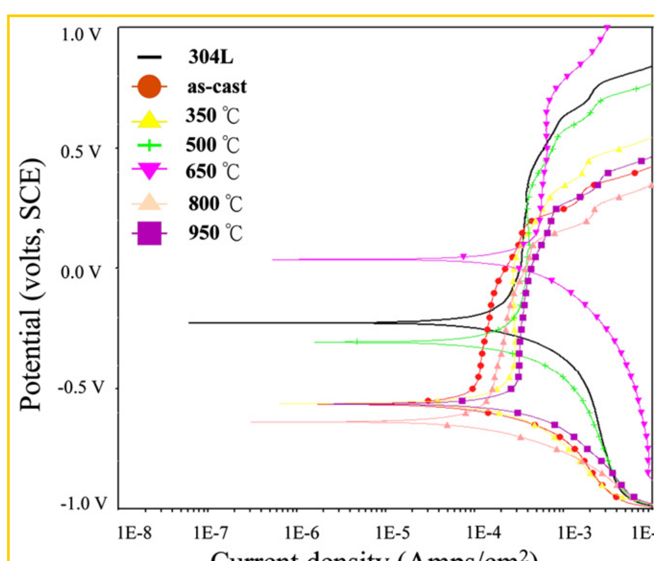

Fig. 5. Potentiodynamic curves of as-cast and aged $Al_{0.5}CoCrFeNi$ alloy specimens in 3.5% NaCl solution at 25 °C.

Table 4
Electrochemical parameters of as-cast and aged $Al_{0.5}CoCrFeNi$ alloy specimens when immersed in 3.5% NaCl solution at 25 °C.

	E_{corr} (V)	i_{corr} (A cm^{-2})	E_{pit} (V)
304 L	−0.22	6.00×10^{-8}	0.48
as-cast	−0.57	1.70×10^{-7}	0.13
350 °C	−0.56	6.40×10^{-6}	0.18
500 °C	−0.30	1.58×10^{-6}	0.28
650 °C	0.04	5.29×10^{-7}	0.71
800 °C	−0.64	3.05×10^{-7}	0.14
950 °C	−0.54	2.22×10^{-6}	0.18

E_{corr}: corrosion potential.
i_{corr}: corrosion current density.
E_{pit}: pitting potential.

temperature. To be specific, the amount of the Al–Ni rich phase decreases as the aging temperature increases, leading to a corresponding increase in the amount of the Al–(Ni, Co, Cr, Fe) phase. These phase changes with the increase in the aging temperature lead to a loss in the corrosion resistance of the $Al_{0.5}CoCrFeNi$ alloy system.

Fig. 6 shows the surface corrosion of fracture that formed following crevice corrosion of the as-cast and aged specimens. The as-cast specimen shows intercrystalline (Al–Ni rich phase) corrosion, as shown in Fig. 6(a). However, all the heat-treated specimens show a few small pits and intercrystalline (Al–Ni rich phase) corrosion on the material surface as well as enhanced crevice corrosion, as shown in Fig. 6(b)–(f), respectively. The surface corrosion on all the specimens indicates that Cl^- ions attack the Al–Ni rich phase sites, both in the as-cast specimen and in all the aged specimens. The images clearly show that the Al–(Ni, Co, Cr, Fe) phases are predominant in the corrosion-resistant $Al_{0.5}CoCrFeNi$ alloy system, suggesting that the precipitates of these phases undergo Cl^- ion attack. The small extent of pitting corrosion is evident on all the specimen surfaces because larger regions of passivation are present on both the as-cast specimen and the specimen aged at 950 °C, which makes it easy for a passive film to be formed over the specimen surface. In addition, intercrystalline (Al–Ni rich phase) corrosion of differing sizes is observed below the surfaces of the specimens aged at a higher temperature as a result of the breakdown of the passive film. Overall, the results shown in Fig. 6 indicate that the properties of the precipitated Al–Ni rich and Al–(Ni, Co, Cr, Fe) phases become significantly unstable as the aging temperature increases and cause the localized corrosion mechanism to change from pitting corrosion to selective corrosion.

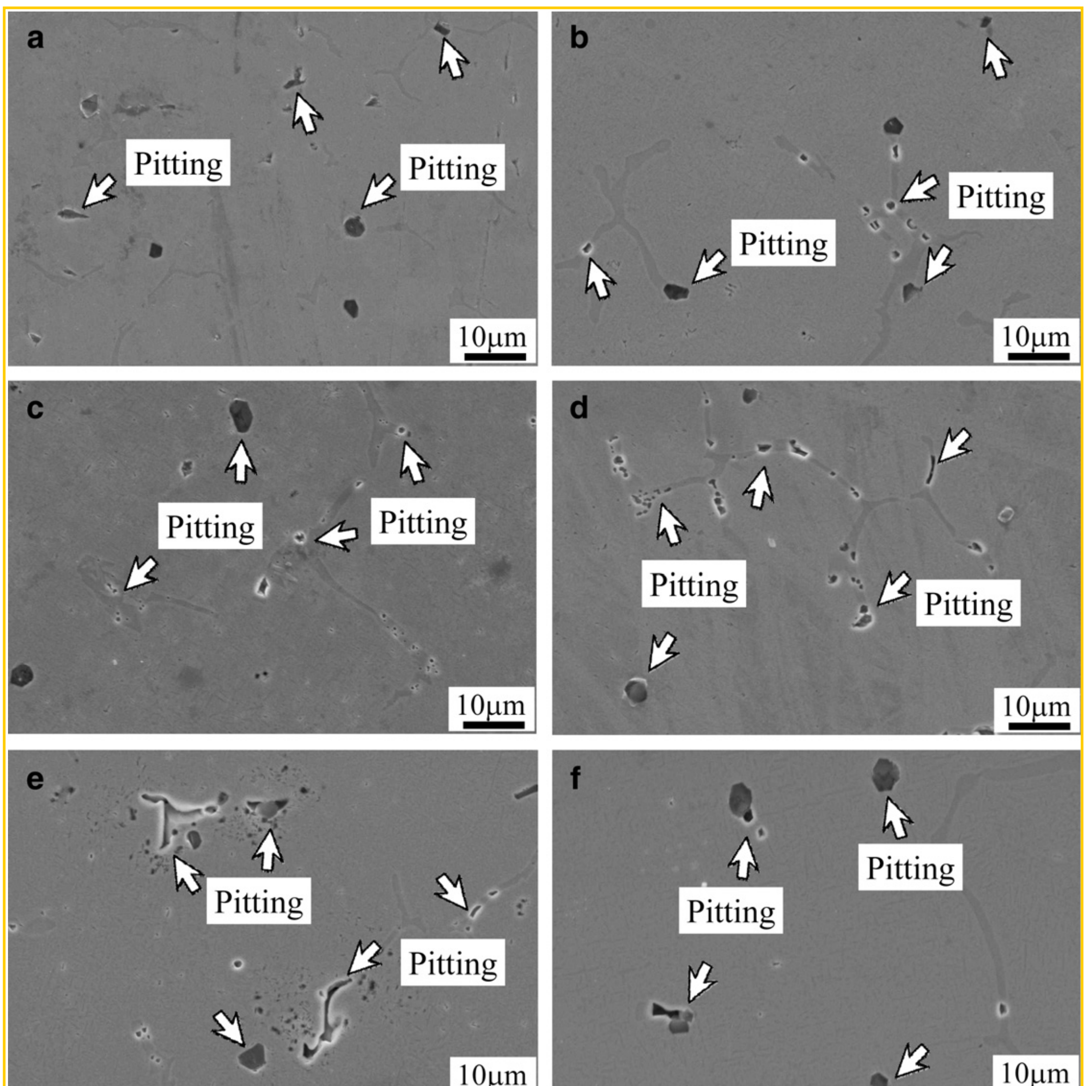

Fig. 6. Surface morphologies of as-cast and aged $Al_{0.5}CoCrFeNi$ alloy specimens following immersion in 3.5% NaCl solution at 25 °C: (a) as-cast, (b) 350 °C, (c) 500 °C, (d) 650 °C, (e) 800 °C, and (f) 950 °C.

4. Conclusions

1. The as-cast high-entropy $Al_{0.5}CoCrFeNi$ alloy has an FCC solid solution structure, whereas the alloys aged at 350–950 °C have an FCC + BCC solid solution structure.
2. The FCC solid solution structure comprises droplet-shaped Al–Ni rich phases and an FCC matrix phase. The duplex FCC + BCC solid solution structure comprises droplet-shaped phases (Al–Ni rich, FCC crystal structure), wall-shaped phases (BCC solid solution structure), needle-shaped phases (Al–(Ni, Co, Cr, Fe), BCC solid solution structure), and the matrix phase.
3. Age hardening of the alloy occurs at all temperatures in the range 350–950 °C. However, the optimal age hardening effect is obtained at temperatures lower than 800 °C.
4. The corrosion potential and pitting potential of the $Al_{0.5}CoCrFeNi$ alloy when immersed in a 3.5 wt% NaCl solution are significantly lower than those of AISI 304L stainless steel. The corrosion properties of the investigated alloy are governed by the aging temperature. The aging of the alloy at temperatures between 350 and 950 °C impairs the pitting resistance of the alloy in chloride environments. In particular, the corrosion properties of the alloy aged at 650 °C differ significantly from those of the other aged specimens. This difference is most probably the result of the microstructure transformation induced by the aging temperature, because the amount of the Al–Ni rich phase decreases as the aging temperature increases, which leads to a corresponding increase in the amount of the Al–(Ni, Co, Cr, Fe) phase.

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